8.6 The Quantum Oscillator

WORKSHEET

The question

STEP III 1988, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q7

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Why should a function like $e^{-\lambda x^2}$ be a sensible guess? What happens to $e^{-\lambda x^2}$ and its derivative as $x \to \pm\infty$?

STAGE 1 The ground state

Covers: the case $n = 0$

Ask yourself Differentiate $y_0 = e^{-\lambda x^2}$ twice and substitute into the equation with $n = 0$. Divide out the exponential.

Pause and try The result must hold for every x, so compare the coefficients of x^2 and the constant terms. Which λ satisfies both? Why must λ be positive?

STAGE 2 The first excited state

Covers: the case $n = 1$

Ask yourself Do the same with $y_1 = xe^{-\lambda x^2}$ and $n = 1$. Does the same value of λ work?

STAGE 3 An identity

Covers: the identity

Ask yourself Differentiate $y_m y_n' - y_n y_m'$ by the product rule. Two terms cancel. Then use the differential equation for y_n'' and for y_m''.

STAGE 4 Orthogonality

Covers: the orthogonality

Pause and try Integrate the identity from $-\infty$ to ∞. What is the left-hand side, given the conditions at infinity? Why can you divide by $2(m-n)\omega$?

What's really going on

Schrödinger's equation

In 1926 Erwin Schrödinger proposed that a quantum particle is described by a *wave function* $\psi(x)$, which for a particle of energy E in a potential $V(x)$ satisfies

$$-\frac{\hbar^2}{2m}\psi'' + V(x)\psi = E\psi.$$

For a spring, $V(x) = \tfrac{1}{2}m\Omega^2 x^2$, and after rescaling x this is exactly the equation in the question, with $E = \left(n + \tfrac{1}{2}\right)\hbar\Omega$. The physical requirement that ψ dies away at infinity (so that the particle is somewhere) can only be met for these energies: that is *quantisation*, falling out of the mathematics. Notice that the lowest energy is not zero but $\tfrac{1}{2}\hbar\Omega$: a quantum oscillator can never be completely at rest. This *zero-point energy* is real and measurable; it is why liquid helium does not freeze at ordinary pressure, even at absolute zero.

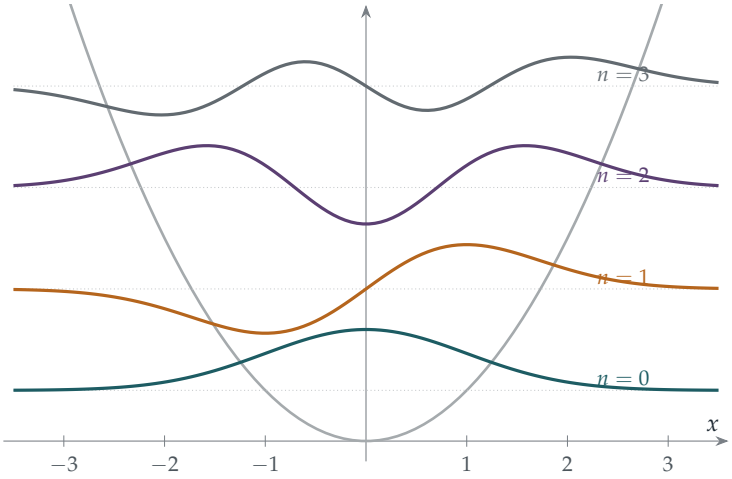


Figure 8.6.1 The quantum harmonic oscillator (with $\omega = 1$). The parabola is the potential energy. Each wave function y_n is drawn at the height of its energy level, $n + \frac{1}{2}$; the levels are equally spaced, and the lowest is above zero. The function y_n crosses zero n times, and any two of them are orthogonal.

Hermite functions and Sturm–Liouville theory

The general solution is $y_n = H_n(\sqrt{\omega}x)e^{-\omega x^2/2}$, where H_n is a polynomial of degree n, the *Hermite polynomial* (Figure 8.6.1). Orthogonality is not special to this equation. Whenever an equation can be written as $(py')' + qy = -\lambda wy$ with suitable boundary conditions, solutions belonging to different values of λ are orthogonal, by exactly the argument of Stage 4. This is *Sturm–Liouville theory* (1830s), and it covers the sine functions of Fourier series (Section 3.7), the vibrations of a drum, the Legendre polynomials of electrostatics and the wave functions of the hydrogen atom. The eigenvalue problem of Section 4.7 is another instance: the critical values found there are the eigenvalues λ, and the orthogonal functions are the natural "modes" of the system.

Try this next

1 **The next state.** Show that $y_2 = (4\lambda x^2 - 1)e^{-\lambda x^2}$, with $\lambda = \frac{\omega}{2}$, satisfies the equation with $n = 2$. Check directly that $\int_{-\infty}^{\infty} y_0 y_2 \, dx = 0$, using $\int_{-\infty}^{\infty} e^{-\omega x^2} \, dx = \sqrt{\pi/\omega}$ and $\int_{-\infty}^{\infty} x^2 e^{-\omega x^2} \, dx = \frac{1}{2\omega}\sqrt{\pi/\omega}$.

2 **No other energies.** Try $y = e^{-\lambda x^2}$ with n replaced by a general number ν. Show that it works only if $\nu = 0$. (In fact, decaying solutions exist only when ν is a non-negative integer.)

3 **Sine waves.** Use the same method to show that $\int_0^{\pi} \sin mx \sin nx \, dx = 0$ for $m \neq n$, starting from $y'' = -n^2 y$ with $y(0) = y(\pi) = 0$.

4 A ladder. Show that if y_n solves the equation, then $y_{n+1} = \omega x y_n - y_n'$ solves it with $n+1$ in place of n. Use this to generate y_1 and y_2 from y_0. (This is the "raising operator" of quantum mechanics.)

Further reading

- **David Griffiths, *Introduction to Quantum Mechanics*** (Cambridge University Press). The standard undergraduate physics text; Chapter 2 treats the harmonic oscillator.
- **Richard Feynman, *QED: The Strange Theory of Light and Matter*** (Princeton University Press). A popular account of quantum mechanics by one of its masters.
- **Mary Boas, *Mathematical Methods in the Physical Sciences*** (Wiley). Covers Hermite polynomials and Sturm–Liouville theory for students of mathematics and physics.

The full guided solution is in Section 8.6:
https://beyondstep.keithmcnulty.org/ch08/quantum-oscillator.html